THE EXT GROUPS OUT OF A FINITELY PRESENTED FUNCTOR ON MODULES VANISH IN POSITIVE DEGREES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The category of module-valued functors on Alg_A has enough injectives. Same chapter, Lemma lemma-enough-injectives, invoked at proof lines 1434-1436. Provides the injective resolution used to compute the Ext groups.
- Ext computed from resolutions, and the Yoneda description of Ext classes. Derived Categories, Lemmas derived-lemma-compute-ext-resolutions and derived-lemma-yoneda-extension, invoked at proof lines 1437-1449. Turns an Ext class into a morphism into the resolution and back into a Yoneda extension.
- Property (*) and its permanence for kernels, images and fibre products of module-valued functors. Same chapter, Lemmas lemma-tangent-injective, lemma-exactness-permanence and lemma-exactness-implies, invoked at proof lines 1456-1496. Makes every functor in the extension satisfy (*) and gives the B-linearity of the comparison maps.
- The tangent functor TF and its multiplication map. Same chapter, Lemma lemma-tangent-functor (item item-mult-map), invoked at proof lines 1489-1496. Supplies the canonical maps F_j(B) -> G_j(B) used in the comparison.

Source reference. Original lemma 058L: sources/stacks-chow/tags.txt, source lines 6784–6784. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original theorem 058K: sources/stacks-chow/tags.txt, source lines 6783–6783. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 06XR: sources/stacks-chow/tags.txt, source lines 8860–8860. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06XU: sources/stacks-chow/tags.txt, source lines 8863–8863. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original reference 06Z9: sources/stacks-adequate/pinned-original-source.tex, source lines 1152–1152. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original reference 06ZC: sources/stacks-adequate/pinned-original-source.tex, source lines 1282–1282. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original reference 06ZD: sources/stacks-adequate/pinned-original-source.tex, source lines 1287–1287. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06Z7: sources/stacks-adequate/pinned-original-source.tex, source lines 1086–1086. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06ZB: sources/stacks-adequate/pinned-original-source.tex, source lines 1271–1271. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06ZF: sources/stacks-adequate/pinned-original-source.tex, source lines 1359–1359. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06Z8: sources/stacks-adequate/pinned-original-source.tex, source lines 1138–1138. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06ZA: sources/stacks-adequate/pinned-original-source.tex, source lines 1195–1195. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring. Let M, P be A-modules with P of finite presentation. Then $\mathrm{Ext}^i_{\mathcal{P}}(\underline{P}, \underline{M}) = 0$ for $i > 0$ where $\mathcal{P}$ is the category of module-valued functors on Alg_A.*

Proof. Choose an injective resolution $\underline{M} \to I^\bullet$ in $\mathcal{P}$, see Lemma 06Z7. By Derived Categories, Lemma 06XR any element of $\mathrm{Ext}^i_{\mathcal{P}}(\underline{P}, \underline{M})$ comes from a morphism $\varphi : \underline{P} \to I^i$ with $d^i \circ \varphi = 0$. We will prove that the Yoneda extension

$$E : 0 \to \underline{M} \to I^0 \to \ldots \to I^{i-1} \times_{\mathrm{Ker}(d^i)} \underline{P} \to \underline{P} \to 0$$

of $\underline{P}$ by $\underline{M}$ associated to φ is trivial, which will prove the lemma by Derived Categories, Lemma 06XU.

For F a module-valued functor on Alg_A say $(*)$ holds if for all $B \in \mathrm{Ob}(Alg_A)$ the functor $TF(B, -)$ on B-modules transforms a short exact sequence of B-modules into a right exact sequence. Recall that the module-valued functors $\underline{M}, I^n, \underline{P}$ each have property $(*)$, see Lemma 06ZA and the remarks preceding it. By splitting $0 \to \underline{M} \to I^\bullet$ into short exact sequences we find that each of the functors $\mathrm{Im}(d^{n-1}) = \mathrm{Ker}(d^n) \subset I^n$ has property $(*)$ by Lemma 06ZF and also that $I^{i-1} \times_{\mathrm{Ker}(d^i)} \underline{P}$ has property $(*)$.

Thus we may assume the Yoneda extension is given as

$$E : 0 \to \underline{M} \to F_{i-1} \to \ldots \to F_0 \to \underline{P} \to 0$$

where each of the module-valued functors F_j has property $(*)$. Set $G_j(B) = TF_j(B, B)$ viewed as a B-module via the *second* B-module structure defined in Lemma 06ZB. Since TF_j is a functor on pairs we see that G_j is a module-valued functor on Alg_A. Moreover, since E is an exact sequence the sequence $G_{j+1} \to G_j \to G_{j-1}$ is exact (see remark preceding Lemma 06ZF). Observe that $T\underline{M}(B, B) = M \otimes_A B = \underline{M}(B)$ and that the two B-module structures agree on this. Thus we obtain a Yoneda extension

$$E' : 0 \to \underline{M} \to G_{i-1} \to \ldots \to G_0 \to \underline{P} \to 0$$

Moreover, the canonical maps

$$F_j(B) = B \otimes_B F_j(B) \longrightarrow TF_j(B, B) = G_j(B)$$

of Lemma 06Z8 (06Z9) are B-linear by Lemma 06ZB (06ZC) and functorial in B. Hence a map

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & \underline{M} & \longrightarrow & F_{i-1} & \longrightarrow & \ldots & \longrightarrow & F_0 & \longrightarrow & \underline{P} & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle 1} & & \downarrow & & & & \downarrow & & \downarrow{\scriptstyle 1} & & \\
0 & \longrightarrow & \underline{M} & \longrightarrow & G_{i-1} & \longrightarrow & \ldots & \longrightarrow & G_0 & \longrightarrow & \underline{P} & \longrightarrow & 0
\end{array}$$

of Yoneda extensions. In particular we see that E and E' have the same class in $\mathrm{Ext}^i_{\mathcal{P}}(\underline{P}, \underline{M})$ by the lemma on Yoneda Exts mentioned above. Finally, let N be a A-module of finite presentation. Then we see that

$$0 \to T\underline{M}(A, N) \to TF_{i-1}(A, N) \to \ldots \to TF_0(A, N) \to T\underline{P}(A, N) \to 0$$

is exact. By Lemma 06ZB (06ZD) with $B = A$ this translates into the exactness of the sequence of A-modules

$$0 \to M \otimes_A N \to G_{i-1}(A) \otimes_A N \to \ldots \to G_0(A) \otimes_A N \to P \otimes_A N \to 0$$

Hence the sequence of A-modules $0 \to M \to G_{i-1}(A) \to \ldots \to G_0(A) \to P \to 0$ is universally exact, in the sense that it remains exact on tensoring with any finitely presented A-module N. Let $K = \mathrm{Ker}(G_0(A) \to P)$ so that we have exact sequences

$$0 \to K \to G_0(A) \to P \to 0 \quad \text{and} \quad G_2(A) \to G_1(A) \to K \to 0$$

Tensoring the second sequence with N we obtain that $K \otimes_A N = \mathrm{Coker}(G_2(A) \otimes_A N \to G_1(A) \otimes_A N)$. Exactness of $G_2(A) \otimes_A N \to G_1(A) \otimes_A N \to G_0(A) \otimes_A N$ then implies that $K \otimes_A N \to G_0(A) \otimes_A N$ is injective. By Algebra, Theorem 058K this means that the A-module extension $0 \to K \to G_0(A) \to P \to 0$ is exact, and because P is assumed of finite presentation this means the sequence is split, see Algebra, Lemma 058L. Any splitting $P \to G_0(A)$ defines a map $\underline{P} \to G_0$ which splits the surjection $G_0 \to \underline{P}$. Thus the Yoneda extension E' is equivalent to the trivial Yoneda extension and we win. $\square$